Pictures Become Numbers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Pictures Become Numbers

> Prepare Pictures Become Numbers: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Encode and reconstruct a small image using an explicit **grid** convention

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Look for: Execute a stated rule including its boundary

Reference example, not the only acceptable wording: At least 5 gives narrow, wide, wide. Greater than 5 gives narrow, narrow, wide. Only the equality case changes among these **input**s.

Input, Process, Output

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

Look for: Trace information through a system without confusing information with a physical action

Reference example, not the only acceptable wording: **Input** includes both request fields and all three catalogue records. The **process** checks science AND pages<=120.

Output contains A only; B fails the length condition and C fails the genre condition. Listing A does not establish that a copy was reserved.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

Concept to explain

A digital image is a **grid** of measured color values.

Term	Meaning
pixel	one picture element in a grid
grid	rows and columns that locate values
channel	one measured color component
resolution	number of pixels used to represent an image

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

The image is binary for transparency, not a claim that ordinary colour images contain only zeros and ones as **pixel** values. Distinguish bit-level storage from the image value convention.

Grid reconstruction is assessed separately from **interpretation**. A learner may name the worked pattern differently while still reconstructing it correctly.

Readiness answer

No. The **fitting** examples may not distinguish the rules, and agreement on those examples does not settle every unseen case.

Squared paper and a dark pencil for each learner

The worked **grid** with its explicit reading convention

Two empty two-row four-column **grid**s for independent assessment

End goal for the pictorial case

By the end, I can encode **pixel** positions and show why a count cannot reconstruct a picture.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Pictures Become Numbers using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT light measured at **grid** locations

2. Process — the deciding step

PROCESS store and transform **pixel** values

3. Output — the result

OUTPUT a displayed, resized, blurred, or classified image

Key idea

CORE IDEA A digital image is a **grid** of measured color values.

How the mechanism works

UNDER THE HOOD Digital images sample a scene across a spatial **grid**. Each **pixel** stores **channel** values, commonly red, green, and blue.

Resolution controls sampling density; transformations operate on numbers, not on an intact object inside the computer.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each **pixel** position.

Step 1

STEP 1 — OBSERVE light measured at **grid** locations

Step 2

STEP 2 — TRACE store and transform **pixel** values

Step 3

STEP 3 — CONCLUDE a displayed, resized, blurred, or classified image

Model answer

MODEL REASONING Each **grid** position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions.

The computer transforms numbers; the human viewer recognizes the shape.

A claim to avoid

DO NOT OVERCLAIM A **pixel** is a measurement at a location, not a tiny copy of the object.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

The teaching case

Draw it, then change the top-left value to 1.

Exact explanation to model

Why the result follows

Initially top-right and bottom-left are black. The change produces three black squares. **Pixel** values encode an image; object recognition is a separate **interpretation**.

Explain the added worked example

Explain the added worked example · 1

Imagine sending a small black-and-white sign using only numbers. We can divide the sign into a **grid**, record 1 for a black square and 0 for a white square, and read each row from left to right, starting at the top.

The receiver needs the width, height and these conventions to reconstruct the same arrangement.

Explain the added worked example · 2

A **pixel** is a position with a recorded value. In our simplified binary image, one value selects black or white. Other image formats can record brightness levels or multiple colour **channel**s.

The convention 1 means black belongs to this exercise; it is not a universal rule for every digital image.

Encoding a picture does not recognise what it depicts. A program can store or display the numbers without calling the image a letter, an arrow or a person.

Interpretation is another operation, just as displaying a book title was different from reading the entire book in Week 2.

The first row is 0,1,0, so only its middle **pixel** is black. The second row is 1,1,1, so all three are black. The third row repeats the first. This reconstructs the displayed cross-like pattern under our convention.

A flattened record would be 0,1,0,1,1,1,0,1,0. Without the **grid** dimensions, the same nine values could also be drawn in one row of nine. Keep the shape information with the values.

Width times height counts **pixel**s; it does not tell us their contents.

Key terms

Term	Meaning
Pixel	One position in a digital image grid with a recorded value or values
Encoding	A convention for representing information with symbols or numbers
Resolution	The number of positions in an image grid, often given as width by height
Spatial arrangement	Where elements are located relative to one another
Interpretation	A meaning or category assigned to a representation using a rule, model or human judgment

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 3

Suppose we discard the row records and retain only black **pixel** count=5. That count matches the cross, but it also matches rows 1,1,1 then 1,1,0 then 0,0,0. The second arrangement fills the top row and two positions below it.

Both have five black **pixel**s and nine **pixel**s in total, yet they are different images.

Stored information	What it establishes	What remains unknown
Width 3 and height 3	There are 9 positions	The value at each position
Black pixel count 5	Five positions are black under this code	Which five positions
Dimensions and all three ordered rows	The full binary grid under this convention	What a viewer or model will call the picture

A **pixel** count can still be a useful feature for some tasks. For example, it can separate empty **grid**s from **grid**s with at least one mark. But it cannot separate every shape.

To judge a feature, ask whether it retains the distinctions required by the task.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Pictures Become Numbers

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

The facts we will use

Supplied facts

Dark=1, light=0; rows are read top to bottom, left to right.

Supplied facts

Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

AI ACADEMY / AI-1/5

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/5

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Pixel convention

Each image has two rows and three columns; 1 means dark.



2. Badge A

Top 101, bottom 010: three dark pixels.



3. Badge B

Top 111, bottom 000: also three dark pixels.



4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Pixel convention	Each image has two rows and three columns; 1 means dark.
2. Badge A	Top 101, bottom 010: three dark pixels.
3. Badge B	Top 111, bottom 000: also three dark pixels.
4. Compare arrangements	The totals agree, but middle-top and middle-bottom positions differ.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption.

What this example does not establish

Keep the stated limit

This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

Change one thing in the pictured case

Change one thing

Change: Change A's bottom-left **pixel** from 0 to 1.

Predicted result

Prediction: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Why the result follows

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Pictures Become Numbers, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

2. State the decision to explain

Write the question to explain: Draw it, then change the top-left value to 1.

3. Reveal the worked reasoning

Reveal after **prediction**s: Initially top-right and bottom-left are black. The change produces three black squares. **Pixel** values encode an image; object recognition is a separate **interpretation**.

Hold back the independent task

Keep for an independent attempt: Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left **pixel** to light. Which number changes?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The two rows are black-white-black and white-black-white, giving three black **pixel**s. Other arrangements share that count, so it cannot recover positions.

For the exit, move one black **pixel** to a previously white position while preserving the total; the image changes but the count does not.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Recall the tied-threshold example. Ask why indistinguishable evidence can leave different answers possible.
7 to 20 minutes	Agree on colour and reading conventions. Reconstruct the worked grid row by row.
20 to 33 minutes	Flatten the values and show why dimensions matter. Compare the two five-black-pixel patterns.

Stage	Teacher action
33 to 45 minutes	Pairs complete W1 and W2, exchanging encodings and reading them back.
Session 2 start 0 to 8 minutes	Show two equal-count grids and ask which input a count-only classifier would receive.
8 to 23 minutes	Complete W3 and diagnose a reversed convention. Require a specific counterexample.
23 to 38 minutes	Collect W4 independently. Check the dimensions and counts before evaluating the explanation.
38 to 45 minutes	Discuss W5 and the exit question. Distinguish stored values from assigned meaning.

Reduce help deliberately

Completion practice

Completion task: A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values. Name the extra information that would distinguish them.

Reasoning to listen for

Teacher check: One row of width 6 is [1,0,0,1,1,0]. Two rows of width 3 are [1,0,0] and [1,1,0]. Specify width and height, plus reading order and colour convention. The values alone do not choose the shape.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Images as numerical **grid**s. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Our convention states that meaning. Other **encoding**s use different scales or **channel**s; read their specification.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Find a position where the original badges differ, then read the changed A back as a picture.

Reasoned teacher answer

The top-middle **pixel** is 0 in A and 1 in B. Changed A is dark at top-left, top-right, bottom-left and bottom-middle.

Changed-case reasoning: A now has four dark **pixel**s; its row **encoding** is 101 / 110. Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Pictures Become Numbers investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Decode a tiny **pixel** picture and predict a blurred version.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR An image is stored as its object name.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The number one always means black: Our convention states that meaning. Other **encoding**s use different scales or **channel**s; read their specification.

A **pixel** count is the picture: Have students reconstruct two different arrangements with the same count.

Storing **pixel**s recognises the object: A **grid** decoder only reconstructs the representation. A separate **interpretation** step assigns a category.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Pictures Become Numbers questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Initially top-right and bottom-left are black. The change produces three black squares. **Pixel** values encode an image; object recognition is a separate **interpretation**.

Independent question

Independent question · withhold the key

Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left **pixel** to light. Which number changes?

Expected independent answer

Teacher key · Expected independent answer

The original light **pixel**s are top-left and bottom-right. The bottom-left entry changes from 0 to 1, giving rows [1,0] and [1,1]. Position and value both matter.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “An image is stored as its object name.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Row-major **encoding** is [1,0,1,0,1,0], with shape 2×3. Interpreting the same list column-first gives top [1,1,1] and bottom [0,0,0]. Values need shape and ordering conventions; **encoding** alone does not recognise an object.

Explanations for the additional workbook examples

W1 Read and reconstruct

Workbook question · W1

Using 1 for black and 0 for white, draw width 3, height 2 with rows 1,0,1 and 0,1,0. Read it back into numbers and count black **pixel**s.

Teacher reasoning · W1

The upper row has black, white, black; the lower row has white, black, white. Reading back gives the supplied two rows. There are three black **pixel**s among six total positions.

W2 Preserve the missing field

Workbook question · W2

A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values. Name the extra information that would distinguish them.

Teacher reasoning · W2

One row of width 6 is [1,0,0,1,1,0]. Two rows of width 3 are [1,0,0] and [1,1,0]. Specify width and height, plus reading order and colour convention. The values alone do not choose the shape.

W3 Test a count feature

Workbook question · W3

A classifier receives only the number of black **pixel**s. Can it always distinguish the chapter's cross from every other three by three pattern? Construct a different pattern with the same count and explain.

Teacher reasoning · W3

No. Rows [1,1,1], [1,1,0], [0,0,0] have five black **pixel**s, as does the cross, but a different arrangement. The classifier receives identical **input** counts for the two, so that feature alone cannot distinguish them.

W4 Design a new counterexample

Workbook question · W4

Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

Teacher reasoning · W4

One valid pair is A [1,1,1,0]/[0,0,0,0] and B [1,0,0,0]/[0,1,0,1]. Both have two rows of four values and three ones. Count retains the number of black positions but discards their locations.

Other valid pairs are accepted if dimensions, counts and arrangements meet the constraints.

W5 Locate an input problem

Workbook question · W5

A program receives the correct **grid** but displays black wherever the code says 0 and white wherever it says 1. A student proposes collecting more classifier examples.

Identify the earlier mismatch and the first check to perform.

Teacher reasoning · W5

The display uses the opposite colour convention from the sender. Check and align the **encoding**/decoding convention, then reconstruct a known **grid**.

More classifier examples would not be the first repair for this identified **input**/display mismatch.

Assess independent transfer

Independent question · withhold the key

Selected case: Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

Selected case answer

Teacher key · Selected case answer

One valid pair is A [1,1,1,0]/[0,0,0,0] and B [1,0,0,0]/[0,1,0,1]. Both have two rows of four values and three ones. Count retains the number of black positions but discards their locations.

Other valid pairs are accepted if dimensions, counts and arrangements meet the constraints.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Pictures Become Numbers to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 6: Sound Becomes a Signal. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide large tactile or printed cells, with numbers as well as shading so colour is unnecessary. Let learners dictate row values.

Extension: construct two **grid**s with equal counts in every row but different arrangements, showing that row totals still lose information.

W4 is secure when both images are two by four, each has three ones, the patterns differ and the explanation identifies lost location information.

If a learner draws valid pictures but mismatches their code, reteach the round-trip check, then assess with fresh patterns.

Week 6 transfers the idea of ordered numerical representations to sound, where the positions refer to times rather than image locations.

Plan the delayed return

Suggested return points in this level: ai-1/6, ai-1/8, ai-1/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Pictures Become Numbers lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Encode and reconstruct a small image using an explicit **grid** convention

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: A message contains 1,0,0,1,1,0 but gives no dimensions. Show a one-row arrangement and a two-row arrangement using the same values.

Name the extra information that would distinguish them.

Supported practice answer

Reasoned practice answer: One row of width 6 is [1,0,0,1,1,0]. Two rows of width 3 are [1,0,0] and [1,1,0]. Specify width and height, plus reading order and colour convention. The values alone do not choose the shape.

Check every learner before discussion: Use the worked case for Images as numerical **grid**s. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Our convention states that meaning. Other **encoding**s use different scales or **channel**s; read their specification.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change A's bottom-left **pixel** from 0 to 1. Require a **prediction** and a reason before comparing. Expected result: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

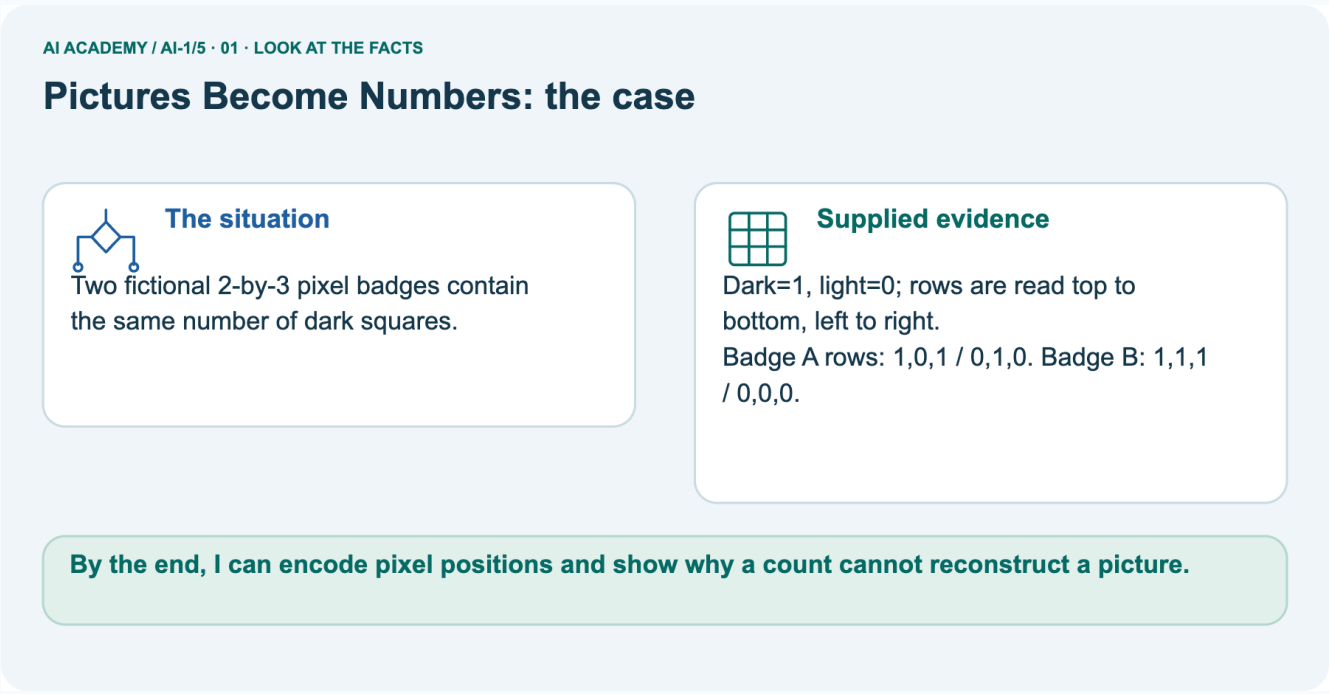
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Pictures Become Numbers graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Pictures Become Numbers: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

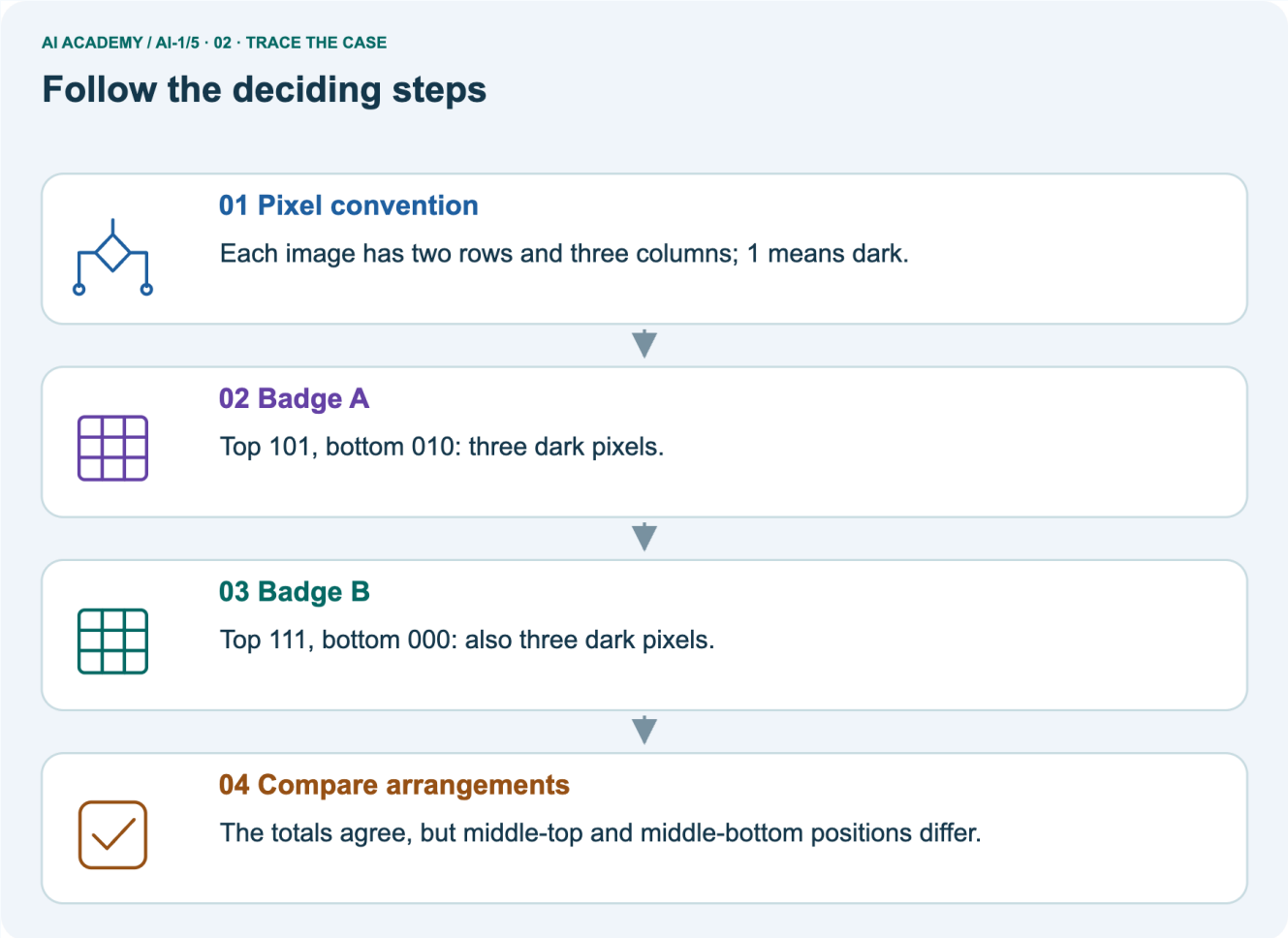
Two fictional 2-by-3 **pixel** badges contain the same number of dark squares. Supplied facts: Dark=1, light=0; rows are read top to bottom, left to right. Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

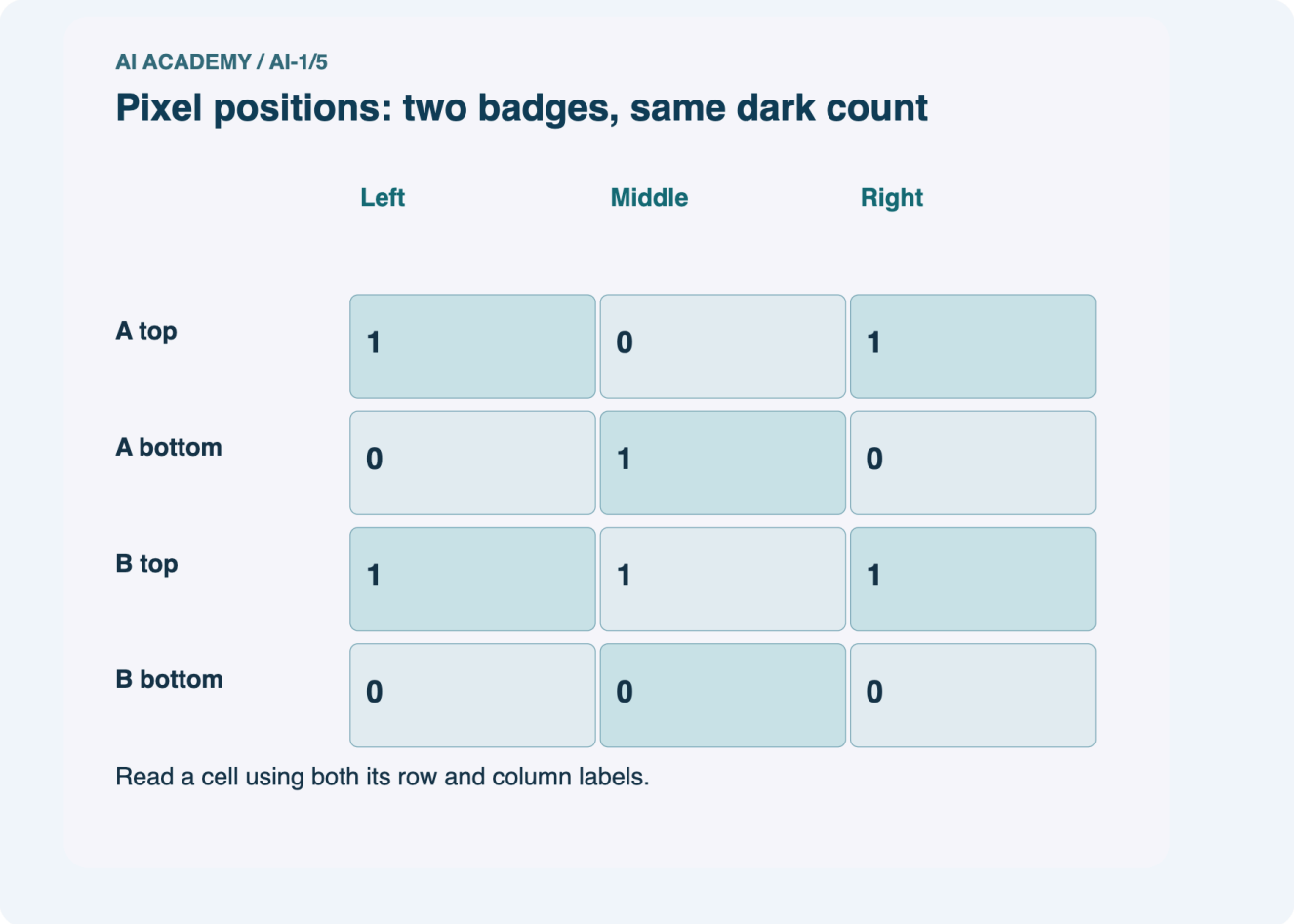
Pixel convention: Each image has two rows and three columns; 1 means dark. → Badge A: Top 101, bottom 010: three dark **pixel**s. → Badge B: Top 111, bottom 000: also three dark **pixel**s. → Compare arrangements: The totals agree, but middle-top and middle-bottom positions differ.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Pixel positions: two badges, same dark count



Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel positions: two badges, same dark count. A top: Left = 1, Middle = 0, Right = 1. A bottom: Left = 0, Middle = 1, Right = 0. B top: Left = 1, Middle = 1, Right = 1. B bottom: Left = 0, Middle = 0, Right = 0.

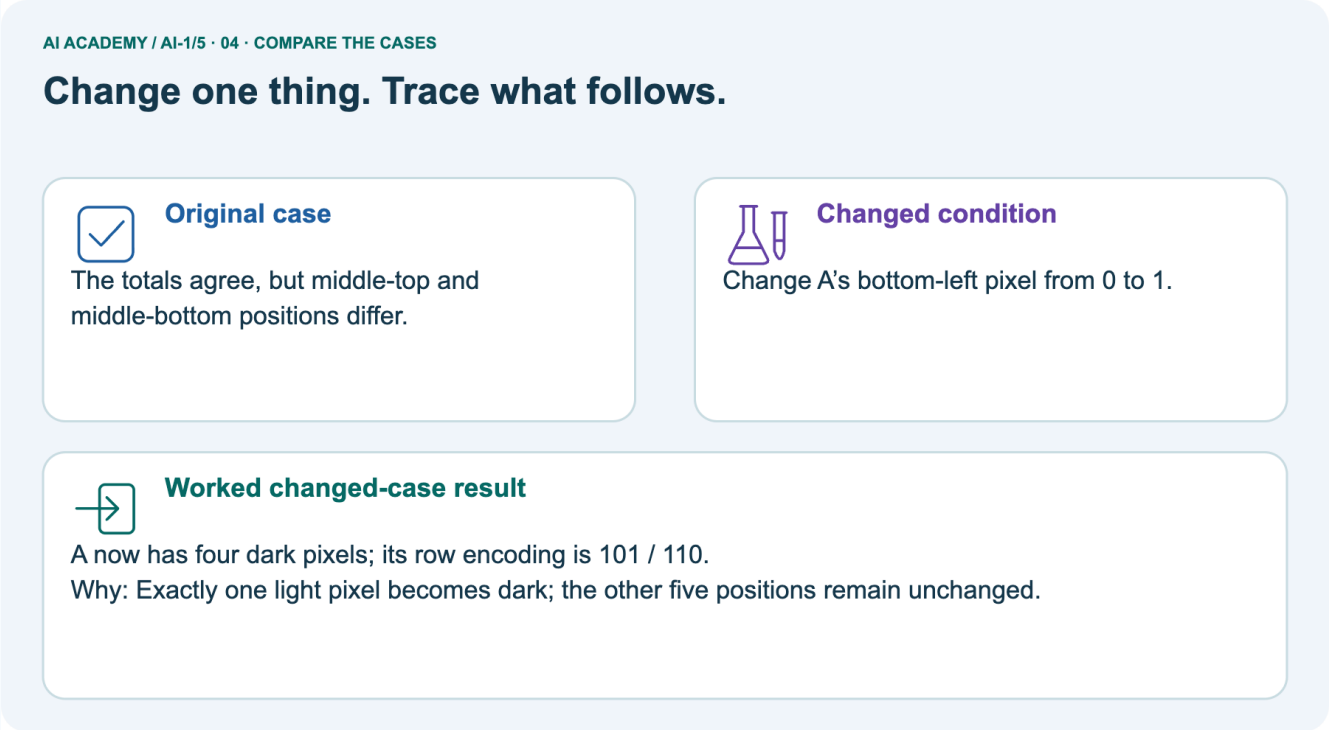
> Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The totals agree, but middle-top and middle-bottom positions differ. Changed condition: Change A's bottom-left **pixel** from 0 to 1. Expected result: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

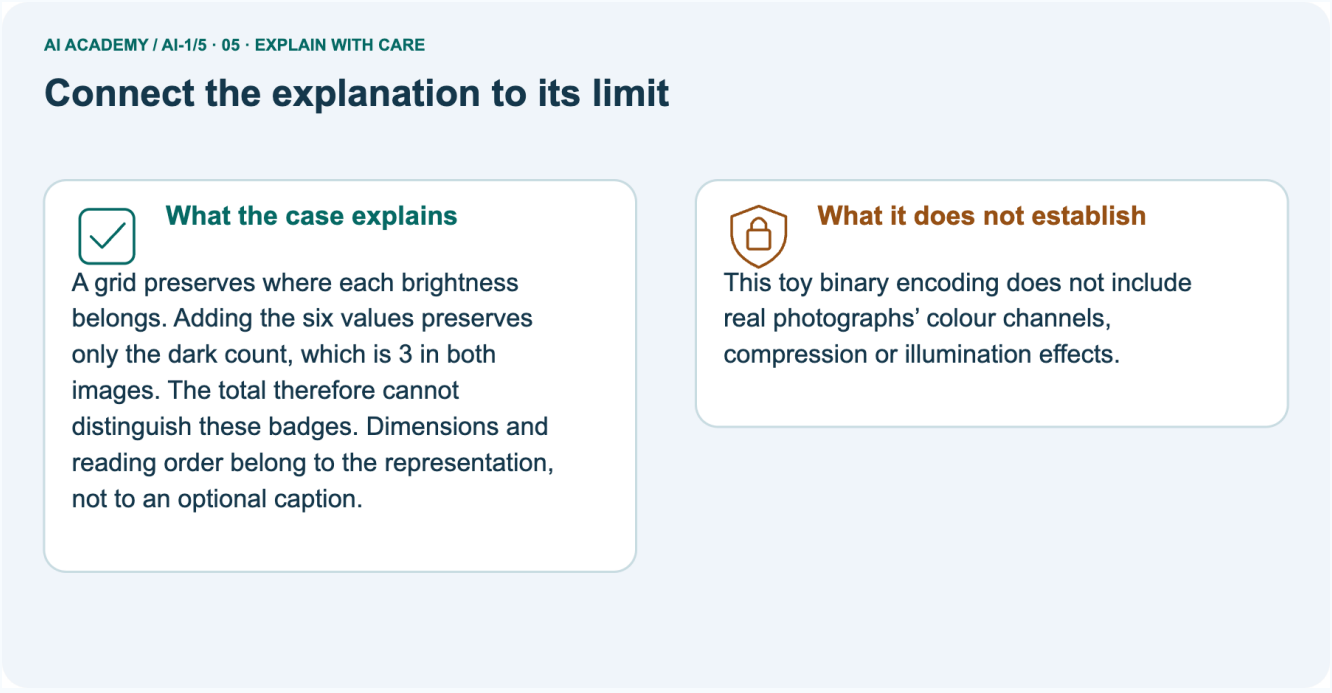
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption. Limit: This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

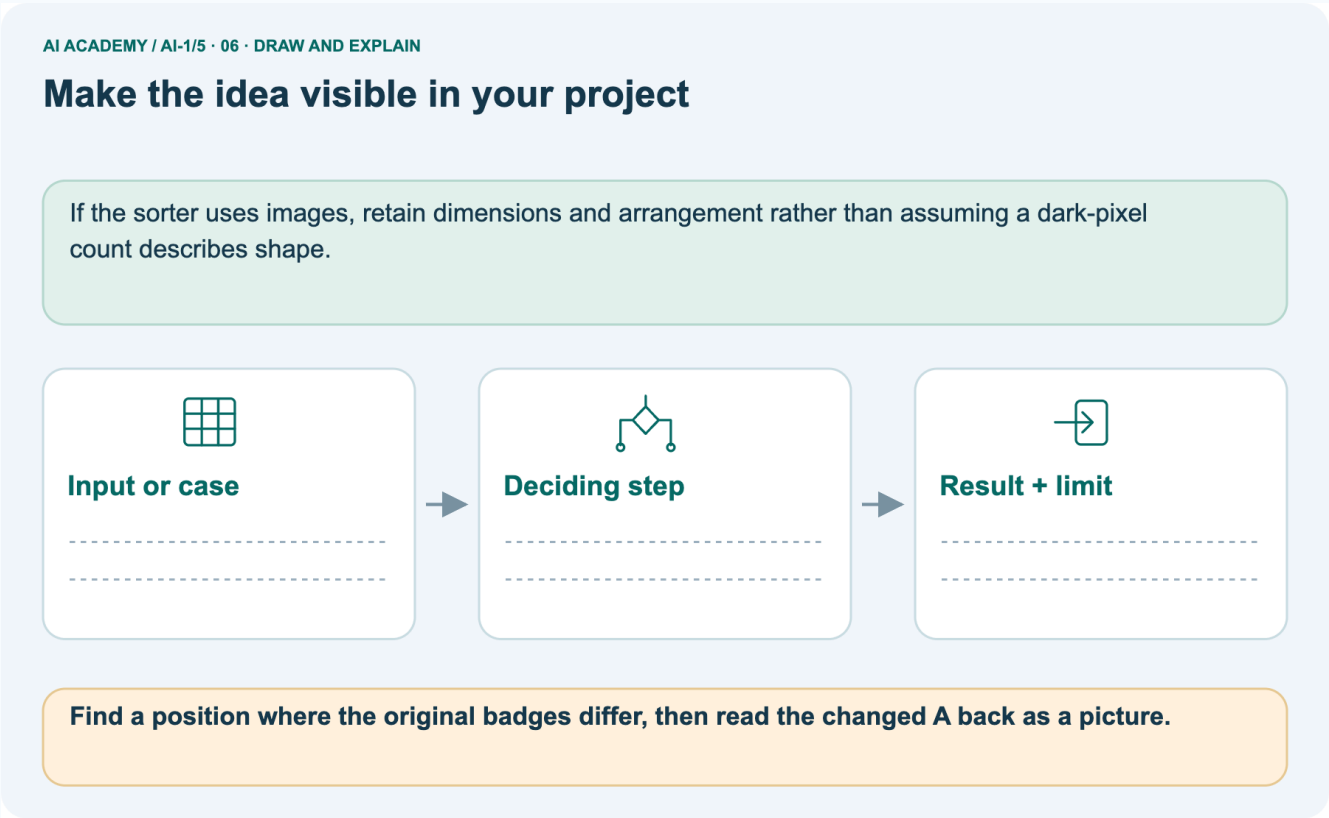
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape. Draw your **input**, deciding step, result and limit.

Supported question: Find a position where the original badges differ, then read the changed A back as a picture.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: The top-middle **pixel** is 0 in A and 1 in B. Changed A is dark at top-left, top-right, bottom-left and bottom-middle.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.